The following are excerpts from this paper here:

https://arxiv.org/abs/astro-ph/0609671

They illustrate the sort of thing I mean when I say "your report should be like the analysis section of a longer paper".

The (astro-)physics of this example don't need to concern us – if you really care, "q" refers to the mass ratio of the two components in a particular class of binary stars; "epsilon" refers to something called the "superhump excess", which basically traces a particular kind of precession in these systems.

"Caliibrators" here just means "data points" for our purposes.

Note that the author here has opted to quote **joint** errors on the two fit parameters. As we will discuss in class, that's definitely one option, but is not required. It is OK if you could errors on "one parameter at a time", so long as you state what you are doing.

Table 7 in P05 provides a list of calibrators for the ϵ–q relation.

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The preferred functional form for the relation is linear, i.e. $q = a + b\epsilon$, with a and b constants. We obtained optimal estimates of a and b by minimizing the χ^2 of the model with respect to the data, allowing explicitly for errors in both q and ϵ and also for the possibility of intrinsic scatter around the relation, σ_{int}. The goodness-of-fit metric then becomes

$$\chi^2 = \sum_{i=1}^{N} \frac{(q - a - b\epsilon)^2}{\sigma_q^2 + b^2\sigma_\epsilon^2 + \sigma_{int}^2}. \quad \text{(A1)}$$

The ϵ–q relation given below is the minimum χ^2 solution. It does not significantly violate the approximate upper limit on BB Dor's mass ratio and achieves a statistically acceptable $\chi^2_\nu = 1.03$ with $\sigma_{int} = 0$. Thus any intrinsic scatter around the calibrating relation must be small compared to the statistical errors on the data points

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The results are shown in Fig. 1. Our preferred fit to the data is given by

$$q(\epsilon) = (0.114 \pm 0.005) + (3.97 \pm 0.41) \times (\epsilon - 0.025). \quad (7)$$

The errors here are 1σ for both parameters jointly, and the shift applied to ϵ ensures that the fit parameters (and their errors) are reasonably uncorrelated. The fit is shown in Fig. 1

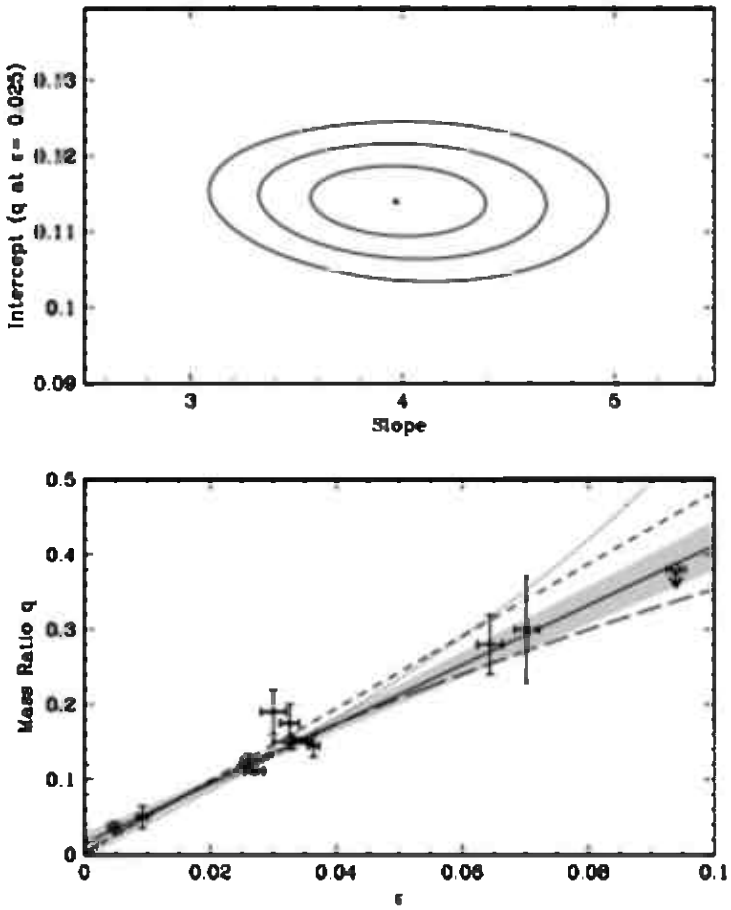


Figure 1. Calibration of the $q - \epsilon$ relation for superhumping CVs. Top panel: constraints on the slope and intercept of the preferred linear calibration. The cross marks the best-fitting parameter combination. The ellipses correspond to 1σ, 2σ and 3σ contours on both parameters jointly. Bottom panel: mass ratio versus period excess for the calibrating stars. The thick solid line is the best-fitting linear calibration, with the shaded area marking the statistical 1σ range. The thick long-dashed line shows P05's calibration, the thin dotted line is Pearson's (2006) calibration, and the thin short-dashed line is the calibration of Goodchild & Ogilvie (2006).